

Multiple Choice (10 marks)

1. Find all zeros in the indicated finite field of the given polynomial with coefficients in that field. $x^3 + 2x + 2$ in $\mathbb{Z}_7$
 - (a) 1
 - (b) 2
 - (c) 2,3
 - (d) 6
2. Using Fermat's theorem, find the remainder of 3^{47} when it is divided by 23.
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
3. Statement 1 | A homomorphism may have an empty kernel. Statement 2 | It is not possible to have a nontrivial homomorphism of some finite group into some infinite group.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Statement 1 | $\mathbb{Q}$ is an extension field of $\mathbb{Z}_2$. Statement 2 | Every non-constant polynomial over a field has a zero in some extension field.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. If $f(z)$ is an analytic function that maps the entire finite complex plane into the real axis, then the imaginary axis must be mapped onto
 - (a) the entire real axis
 - (b) a point
 - (c) a ray
 - (d) an open finite interval

True / False (10 marks)

Answer True or False

6. True or False: The order of a finite group G containing a subgroup of order five and no elements other than the identity as their own inverses can be 35.
7. True or False: The polynomial $8x^3 + 6x^2 - 9x + 24$ satisfies the Eisenstein criterion for irreducibility over $\mathbb{Q}$ with $p=5$.
8. True or False: If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x) + g(x)) = \deg(f(x)) + \deg(g(x))$.
9. True or False: The degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$ is 4.
10. True or False: Every field is also a ring, as fields satisfy all the properties required for a ring.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the validity of the following statements: (1) If H and K are subgroups of a group G , then $|HK| = |H||K|/|H \cap K|$, and (2) A group of order $2p$, where p is an odd prime, is either cyclic or isomorphic to D_{2p} . Provide detailed reasoning and examples to support your analysis.
12. Given the characteristic polynomial $\det(A - I) = -\lambda^3 + 3\lambda^2 - 3\lambda + 1$ for a 3×3 matrix A , analyze the implications.

End of Paper